

SIMATS ENGINEERING

Department of Computer Science & Engineering

CO4-AT1 - Industrial Problem Solving | Programming in Java

Student Name	K. V. S. S. Jeswanth Babu
Register No.	192472012
Course Code	CSA0904
Course Name	Programming in Java
Assignment	CO4-AT1 - Industrial Problem Solving

QUESTION 1: MULTI-WINDOW GUI TICKET BOOKING SYSTEM

Problem Statement:

Develop a multi-window GUI application for ticket booking with features like seat selection, booking confirmation, and cancellation. Use Swing components such as buttons, panels, and menus along with appropriate layout managers to design the interface. Implement event handling to manage user interactions and ensure accurate seat allocation. Validate user inputs and handle exceptions during booking. Ensure smooth navigation between windows.

SOURCE CODE

```
import java.awt.*;
import javax.swing.*;
import java.util.HashMap;
import java.util.Map;

// Main Public Class at the Top
public class TicketBookingSystem extends JFrame {
    private static final int ROWS = 4;
    private static final int COLS = 5;

    // Shared Data Store
    private final JButton[][] seatButtons = new JButton[ROWS][COLS];
    private final boolean[][] seatOccupied = new boolean[ROWS][COLS];
    private final Map<String, Ticket> activeBookings = new HashMap<>();

    private String selectedSeat = null;
    private final JLabel lblSelectedSeat;
    private final JTextField txtPassengerName;
    private final JComboBox<String> cmbShow;

    public TicketBookingSystem() {
        setTitle("Cinema & Event Ticket Booking System");
        setSize(750, 550);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setLayout(new BorderLayout(10, 10));

        // Menu Bar Setup
        setJMenuBar(createMenuBar());

        // Header Panel
        JPanel pnlHeader = new JPanel();
```

```

pnlHeader.setBackground(new Color(30, 60, 114));
JLabel lblHeader = new JLabel("SEAT SELECTION & RESERVATION PORTAL");
lblHeader.setForeground(Color.WHITE);
lblHeader.setFont(new Font("Arial", Font.BOLD, 18));
pnlHeader.add(lblHeader);
add(pnlHeader, BorderLayout.NORTH);

// Center Panel: Screen & Seat Grid
JPanel pnlCenter = new JPanel(new BorderLayout(10, 10));
pnlCenter.setBorder(BorderFactory.createEmptyBorder(10, 20, 10, 20));

JLabel lblScreen = new JLabel("--- SCREEN / STAGE THIS WAY ---", SwingConstants.CENTER);
lblScreen.setOpaque(true);
lblScreen.setBackground(new Color(220, 220, 220));
lblScreen.setFont(new Font("SansSerif", Font.BOLD, 12));
lblScreen.setBorder(BorderFactory.createLineBorder(Color.GRAY));
pnlCenter.add(lblScreen, BorderLayout.NORTH);

JPanel pnlSeats = new JPanel(new GridLayout(ROWS, COLS, 10, 10));
char rowChar = 'A';
for (int r = 0; r < ROWS; r++) {
    for (int c = 0; c < COLS; c++) {
        String seatLabel = "" + (char)(rowChar + r) + (c + 1);
        JButton btn = new JButton(seatLabel);
        btn.setBackground(new Color(178, 222, 39));
        btn.setFont(new Font("Arial", Font.BOLD, 12));
        final int rowIdx = r;
        final int colIdx = c;

        btn.addActionListener(e -> selectSeat(seatLabel, rowIdx, colIdx));
        seatButtons[r][c] = btn;
        pnlSeats.add(btn);
    }
}
pnlCenter.add(pnlSeats, BorderLayout.CENTER);
add(pnlCenter, BorderLayout.CENTER);

// Right Panel: Booking Form
JPanel pnlForm = new JPanel();
pnlForm.setLayout(new BoxLayout(pnlForm, BoxLayout.Y_AXIS));
pnlForm.setBorder(BorderFactory.createTitledBorder("Passenger Details"));
pnlForm.setPreferredSize(new Dimension(280, 0));

pnlForm.add(new JLabel("Select Show:"));
cmbShow = new JComboBox<>(new String[]{"Interstellar (Evening)", "Inception (Night)", "Oppenheimer (Matinee)"});
pnlForm.add(cmbShow);
pnlForm.add(Box.createVerticalStrut(10));

pnlForm.add(new JLabel("Passenger Name:"));
txtPassengerName = new JTextField(15);
pnlForm.add(txtPassengerName);
pnlForm.add(Box.createVerticalStrut(10));

pnlForm.add(new JLabel("Selected Seat:"));
lblSelectedSeat = new JLabel("None");
lblSelectedSeat.setFont(new Font("Arial", Font.BOLD, 14));
lblSelectedSeat.setForeground(new Color(30, 60, 114));
pnlForm.add(lblSelectedSeat);
pnlForm.add(Box.createVerticalStrut(20));

JButton btnConfirm = new JButton("Confirm Booking");
btnConfirm.setBackground(new Color(30, 60, 114));
btnConfirm.setForeground(Color.WHITE);
btnConfirm.setFocusPainted(false);
btnConfirm.addActionListener(e -> processBooking());

```

```

        pnlForm.add(btnConfirm);

        add(pnlForm, BorderLayout.EAST);

        // Status Footer
        JLabel lblFooter = new JLabel(" Green: Available | Red: Booked | Yellow: Selected", SwingConstants.CENTER);
        lblFooter.setBorder(BorderFactory.createEmptyBorder(5, 5, 5, 5));
        add(lblFooter, BorderLayout.SOUTH);
    }

    private JMenuBar createMenuBar() {
        JMenuBar menuBar = new JMenuBar();
        JMenu menuOptions = new JMenu("Options");

        JMenuItem itemCancel = new JMenuItem("Cancel Ticket");
        itemCancel.addActionListener(e -> new CancellationDialog(this).setVisible(true));

        JMenuItem itemReset = new JMenuItem("Reset All");
        itemReset.addActionListener(e -> resetAllSeats());

        JMenuItem itemExit = new JMenuItem("Exit");
        itemExit.addActionListener(e -> System.exit(0));

        menuOptions.add(itemCancel);
        menuOptions.add(itemReset);
        menuOptions.addSeparator();
        menuOptions.add(itemExit);
        menuBar.add(menuOptions);
        return menuBar;
    }

    private void selectSeat(String seatLabel, int r, int c) {
        if (seatOccupied[r][c]) {
            JOptionPane.showMessageDialog(this, "Seat " + seatLabel + " is already booked!", "Seat Unavailable", JOptionPane.WARNING_MESSAGE);
            return;
        }

        for (int i = 0; i < ROWS; i++) {
            for (int j = 0; j < COLS; j++) {
                if (!seatOccupied[i][j]) {
                    seatButtons[i][j].setBackground(new Color(178, 222, 39));
                }
            }
        }

        selectedSeat = seatLabel;
        lblSelectedSeat.setText(selectedSeat);
        seatButtons[r][c].setBackground(Color.YELLOW);
    }

    private void processBooking() {
        try {
            String name = txtPassengerName.getText().trim();
            String show = (String) cmbShow.getSelectedItem();

            if (name.isEmpty()) {
                throw new InvalidBookingException("Passenger name cannot be empty.");
            }
            if (!name.matches("^[a-zA-Z\\s]+$")) {
                throw new InvalidBookingException("Name must contain only alphabets and spaces.");
            }
            if (selectedSeat == null) {
                throw new InvalidBookingException("Please select an available seat first.");
            }
        }
    }

```

```

String bookingId = "BK" + (1000 + activeBookings.size() + 1);
Ticket ticket = new Ticket(bookingId, name, selectedSeat, show);
activeBookings.put(bookingId, ticket);

int r = selectedSeat.charAt(0) - 'A';
int c = Character.getNumericValue(selectedSeat.charAt(1)) - 1;
seatOccupied[r][c] = true;
seatButtons[r][c].setBackground(new Color(231, 76, 60));
seatButtons[r][c].setEnabled(false);

selectedSeat = null;
lblSelectedSeat.setText("None");
txtPassengerName.setText("");

new BookingConfirmationDialog(this, ticket).setVisible(true);

} catch (InvalidBookingException ex) {
    JOptionPane.showMessageDialog(this, ex.getMessage(), "Input Validation Error", JOptionPane.ERROR_MESSAGE);
} catch (Exception ex) {
    JOptionPane.showMessageDialog(this, "An unexpected error occurred: " + ex.getMessage(), "Error", JOptionPane.ERROR_MESSAGE);
}
}

public boolean cancelBooking(String bookingId) {
    if (!activeBookings.containsKey(bookingId)) {
        return false;
    }
    Ticket t = activeBookings.remove(bookingId);
    String seat = t.getSeatNumber();
    int r = seat.charAt(0) - 'A';
    int c = Character.getNumericValue(seat.charAt(1)) - 1;
    seatOccupied[r][c] = false;
    seatButtons[r][c].setBackground(new Color(178, 222, 39));
    seatButtons[r][c].setEnabled(true);
    return true;
}

private void resetAllSeats() {
    activeBookings.clear();
    selectedSeat = null;
    lblSelectedSeat.setText("None");
    txtPassengerName.setText("");
    for (int i = 0; i < ROWS; i++) {
        for (int j = 0; j < COLS; j++) {
            seatOccupied[i][j] = false;
            seatButtons[i][j].setBackground(new Color(178, 222, 39));
            seatButtons[i][j].setEnabled(true);
        }
    }
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> new TicketBookingSystem().setVisible(true));
}

// Custom Exception Class
class InvalidBookingException extends Exception {
    public InvalidBookingException(String message) {
        super(message);
    }
}

// Model Class representing a Ticket

```

```

class Ticket {
    private final String bookingId;
    private final String passengerName;
    private final String seatNumber;
    private final String movieOrEvent;

    public Ticket(String bookingId, String passengerName, String seatNumber, String movieOrEvent) {
        this.bookingId = bookingId;
        this.passengerName = passengerName;
        this.seatNumber = seatNumber;
        this.movieOrEvent = movieOrEvent;
    }

    public String getBookingId() { return bookingId; }
    public String getPassengerName() { return passengerName; }
    public String getSeatNumber() { return seatNumber; }
    public String getMovieOrEvent() { return movieOrEvent; }
}

// Window 2: Booking Confirmation Dialog
class BookingConfirmationDialog extends JDialog {
    public BookingConfirmationDialog(JFrame parent, Ticket ticket) {
        super(parent, "Booking Confirmed", true);
        setSize(380, 320);
        setLocationRelativeTo(parent);
        setLayout(new BorderLayout(10, 10));

        JPanel pnlHeader = new JPanel();
        pnlHeader.setBackground(new Color(46, 204, 113));
        JLabel lblTitle = new JLabel("TICKET CONFIRMED");
        lblTitle.setFont(new Font("Arial", Font.BOLD, 16));
        lblTitle.setForeground(Color.WHITE);
        pnlHeader.add(lblTitle);
        add(pnlHeader, BorderLayout.NORTH);

        JPanel pnlBody = new JPanel(new GridLayout(4, 2, 8, 8));
        pnlBody.setBorder(BorderFactory.createEmptyBorder(20, 25, 20, 25));

        pnlBody.add(new JLabel("Booking ID:"));
        pnlBody.add(new JLabel(ticket.getBookingId()));

        pnlBody.add(new JLabel("Passenger Name:"));
        pnlBody.add(new JLabel(ticket.getPassengerName()));

        pnlBody.add(new JLabel("Show/Event:"));
        pnlBody.add(new JLabel(ticket.getMovieOrEvent()));

        pnlBody.add(new JLabel("Seat Number:"));
        pnlBody.add(new JLabel(ticket.getSeatNumber()));

        add(pnlBody, BorderLayout.CENTER);

        JButton btnClose = new JButton("Done");
        btnClose.addActionListener(e -> dispose());
        JPanel pnlBottom = new JPanel();
        pnlBottom.add(btnClose);
        add(pnlBottom, BorderLayout.SOUTH);
    }
}

// Window 3: Ticket Cancellation Dialog
class CancellationDialog extends JDialog {
    public CancellationDialog(TicketBookingSystem parent) {
        super(parent, "Cancel Ticket", true);
        setSize(360, 200);
        setLocationRelativeTo(parent);
    }
}

```

```

setLayout(new BorderLayout(10, 10));

JPanel pnlBody = new JPanel(new GridLayout(2, 2, 10, 10));
pnlBody.setBorder(BorderFactory.createEmptyBorder(20, 20, 10, 20));

pnlBody.add(new JLabel("Enter Booking ID:"));
JTextField txtBookingId = new JTextField();
pnlBody.add(txtBookingId);

add(pnlBody, BorderLayout.CENTER);

JPanel pnlButtons = new JPanel();
JButton btnCancel = new JButton("Cancel Reservation");
JButton btnClose = new JButton("Close");

btnCancel.setBackground(new Color(231, 76, 60));
btnCancel.setForeground(Color.WHITE);

btnCancel.addActionListener(e -> {
    String bId = txtBookingId.getText().trim();
    if (bId.isEmpty()) {
        JOptionPane.showMessageDialog(this, "Please enter a valid Booking ID.", "Error", JOptionPane.ERROR_MESSAGE);
        return;
    }
    boolean success = parent.cancelBooking(bId);
    if (success) {
        JOptionPane.showMessageDialog(this, "Booking " + bId + " successfully cancelled and seat released.", "Success", JOptionPane.INFORMATION_MESSAGE);
        dispose();
    } else {
        JOptionPane.showMessageDialog(this, "Booking ID not found in active records.", "Not Found", JOptionPane.WARNING_MESSAGE);
    }
});

btnClose.addActionListener(e -> dispose());
pnlButtons.add(btnCancel);
pnlButtons.add(btnClose);
add(pnlButtons, BorderLayout.SOUTH);
}

```

OUTPUT SCREENSHOT

